

1. Funds & Backtest Design

Effortless Invest offers nine funds. It is built by using three portfolio construction methods across three fund groups defined by investable universe rather than by strategy. The multi asset group includes all 60 assets of 50 equities and 10 cryptocurrencies, allowing the optimiser to allocate freely across both assets simultaneously. The equity group is restricted to the 50 equities only with zero crypto exposure and the crypto group is restricted to only 10 cryptocurrencies with zero equity exposure. The same three optimisation methods are then applied within each universe which results in the nine funds to raise a specific question: is an investor better off spreading their money across both equities and crypto, or is sticking to one better?

Optimisation methods. All three methods are solved subject to the same constraints. The weights must be non-negative and sum to one. This means short selling isn't allowed:

1. Minimum variance selects weights to minimise the portfolio variance based on the covariance matrix of trailing returns. It does not include expected returns at all. This makes the method the most defensive one, but it can still end up overweighting assets that are low volatility and weakly correlated with others. Even if those same assets have weak expected returns.
2. Maximum sharpe selects weights to maximise the expected portfolio return ratio to portfolio volatility. This is the most estimation sensitive method as it optimises directly on expected returns.
3. Risk parity aims for each asset to contribute equally to the overall portfolio risk. A small minimum weight will be imposed on each asset, this is to avoid instability when an asset's risk contribution approaches zero.

Walk forward backtest, this means that all the funds are backtested out of the sample. At each monthly rebalance date (first trading day of the month), weights are solved using the trailing 252 day window of returns before that date. Also, the current day's return is excluded from the estimation window to prevent look ahead bias. Once solved, the weights are held constant and applied to realised return until the next rebalance. Sharpe ratios are calculated using a risk free rate of zero at every rebalance, per the brief's guideline. Trading costs are also assumed to be zero at every rebalance. This means that real world returns would be somewhat lower than reported here.

The 252 day window was chosen because it reflects roughly a year of equity trading, balancing stable covariance estimates against responsiveness to recent market conditions. This is because shorter windows are noisier, but longer ones are slower to adapt but more stable. The monthly rebalancing satisfies the brief's minimum requirement, as it matches real world practice and keeps the 3 year backtest to 36 rebalances without excessive turnover. Also, neither the window length nor rebalancing frequency was actually tested against other options as they're both just starting points not values I proved were optimal. Testing different window lengths would be a good next step to check how much this choice actually matters.

Annualised and calendar alignment. Following the brief's requirement, the combined and equity fund groups are annualised using 252 trading days, and the crypto only group uses 365. This is not just because crypto trades seven days a week but rather it reflects on how

each fund's return series was constructed. Also, the combined fund's return panel is built using left merging crypto prices onto the calendar of equity trading. Therefore the combined fund observes and rebalances on equity trading days, where 252 is the correct annualisation factor for that observed panel. The crypto only fund retains its 365 days calendar and is annualised accordingly.

Limitations. Beyond what's already stated, transaction costs or slippage was not included. This means that the results are more likely to be more optimistic than what a real fund would achieve. This is because it hits max sharpe and risk parity harder than min variance, as their weights tend to move around more at each rebalance. Max sharpe also has a weakness: it heavily leans on expected returns estimated from 252 days of noisy data, where it's a small sample to build return forecasts from. Therefore its result should not be read as evidence that the method is the best, as some of it may just reflect noise in the estimate.

2. Out of Sample Results & Fact Sheet

All nine funds were backtested out of sample from 2021-01-04 to 2023-12-29, with 2020 used as the 252 day estimation burn in

Table 1. Fund performance out of sample, 2021-01-04 - 2023-12-29. Returns and volatility are annualised at 252 days for the combined and equity funds, 365 days for crypto. Max drawdown is the biggest peak to trough loss over this period.

Diversification Benefit. The benefit of adding crypto to the equity doesn't help evenly, it depends on the method. For max sharpe, the benefit is clear. The combined score of 1.179 is nearly double compared to equity only of 0.623. However for min variance and risk parity, equity only does better (0.676 vs 0.512, and 0.874 vs 0.828). This actually makes sense, because max sharpe is chasing returns, making use of crypto's big (if volatile) upside. Min variance and risk parity care more about stability, so they are better off just sticking with equity only. In hindsight, diversification into crypto is only better when the method is actually built to chase returns in the first place.

Crypto Funds: High Sharpe, High Cost. Crypto min variance has the highest sharpe and returns out of the nine funds (2.164 and 146.7%). This seems odd for a method that is supposed to be 'defensive'. However, looking at what the fund actually held explains why. Weights concentrated in BTC-USD (48.5%) and TRX-USD (24.0%). Min variance finds the one that moves around the least, and doesn't move in the same direction as everything else. It correctly picked the most liquid, lowest volatility crypto asset and kept it. This excessive returns shows that even this 'defensive' segment of the market saw a big rally in 2021, which is not a flaw in the method. Why min variance settled on a 24% TRX allocation is less clear and deserves a closer look than this report can give. However, every crypto fund shows the steepest drawdowns of the nine funds (-67.5% to -89.3%) which lines up with the 2022 crash. That's a level of loss that any investor would not invest if they can't handle losing most of their capital.

Equity method comparison. Among equity only funds, risk parity (Sharpe 0.874) outperforms min variance (0.676) and max sharpe (0.623) in terms of risk adjusted performance. This is notable because risk parity is often used as the secondary method.

However in this case, spreading the risk across evenly beats chasing forecasted returns and minimising variance.

Fund Fact Sheet. As seen in the Streamlit app's fund tab: a growth of \$1 chart, the same four performance metrics shown as individual summary cards, and top 10 holdings by weight. Combine min variance is led by WMT, MRK, ABBV, GILD, AMGN, leaning towards safer and low volatility large caps, as expected of a fund built to minimise risk.

Figure 1 shows that, within the combined group, max sharpe ends at \$1.96, risk parity at \$1.52, and min variance at \$1.20. With all starting from \$1 on 2021-01-04. This confirms the annualised return ranking in table 1 while showing the higher return funds took a rough path

Drawdown risk. Figure 2 shows a shallow and repeated -5% to -15% drawdown through 2021. After that, the fund entered a deeper and sustained decline from May 2022, reaching a low of -26.28% on 30 September 2022 alongside the broader selloff. It shows that recovery is slow, the fund spends most of 2023 below its earlier high. Any investor holding through 2022-2023 would have been underwater for well over a year, which is something that the Sharpe ratio alone would not show.

Weights over time. Figure 3 shows the named tickers started at 61.9% of the fund in Jan 2021, it then dropped to a low of 20.3% by Dec 2021, while 'Other' growing to 79.7%. The weights shifted back towards the named tickers where it reached a second peak of 49.9% by Apr 2023, before falling again to 16.7% by Dec 2023 (lowest point in the sample), and 'Other' reached its overall maximum of 83.3%. The reason behind its fluctuation is due to the model not picking a fixed set of holdings, every month it looks back at the last year of returns and resolves for whatever mix is lowest risk right now. If a different set of assets looks calmer than the named large caps did last year, the weights shift there instead. Therefore the allocation tracks whatever happens to the least volatile over the last 252 days.

Sharpe comparison across funds. Figure 4 shows the spread within crypto clearly: crypto min variance is the tallest bar at 2.164, and crypto max sharpe is the shortest at 0.323. This shows that the same asset class produces both the best and worst result in the whole lineup. That is compared to equity, where all three methods result in a narrow band of values between 0.623 and 0.874. That difference tells us about crypto's extreme volatility. A strategy based on noisy expected-return estimates, such as max-Sharpe, gets punished, whereas a strategy like min-variance, that simply skips over all that noise by selecting the quietest asset (mostly BTC), is rewarded and comes across as looking great by comparison. Outside of crypto, combined_max_sharpe is the best performer at 1.179 ahead of all other combined or equity funds.

3. Sentiment Index

Effortless Invest has created an index of news sentiment scores for the 10 equity sectors, providing investors with a perspective on market sentiment that is separate from price data. The sentiment model scores 146,830 headlines covering 10 sectors multiplied by 1,005 trading days, using VADER as it is (no extension of VADER lexicon for finance).

Methodology. Each headline is scored on the raw text of its title, including capitalisation and punctuation. This is significant because the casing and punctuation are important to detect

negation and emphasis. For example, capitalisation and exclamation marks are important signals to VADER, not noise to be stripped. Headline level scores are averaged to a ticker day score, then weighted across tickers within each sector to produce a sector day sentiment value. A ticker day with no headline will be treated as neutral (0.0), rather than dropped or carried forward from prior day's sentiment. The absence of coverage is not evidence of positive or negative sentiment. In order to prevent look ahead bias, the sentiment signal is one trading day behind so that a headline published on Monday is first usable on Tuesday, and a headline published on the weekends are mapped to the trading day that follows which is then delayed by another trading day so it's usable on Tuesday.

Validation & Distribution. The overall sentiment of all scored headlines is close to neutral. 48.8% are exactly neutral (compound score of 0.0), the mean is 0.106, and the median is 0.0 (which is no sentiment at all under plain VADER). The range of scores is from -0.92 to 0.96 with a 25th percentile score of 0.0 and 75th percentile score of 0.296. This indicates that the distribution is clustered around the neutral value and has a slight positive skew.

This 48.8% is close to what the brief expects from plain VADER on financial headlines. This does not mean VADER is failing but rather its working as it is designed to. It's a generic sentiment model, and the finance headlines use specialised language that VADER was not trained to recognise. A headline like 'earnings beat estimates' carries real sentiment to a financial reader, but for VADER a true zero and false neutral look the same. Therefore telling them apart would need more work, like adding a finance specific lexicon.

Sector Patterns. Figure 5 shows the equity sector day sentiment for all ten equity sectors after being smoothed for readability using a 20 day moving average. The average sentiment is positive in each sector, and on average, none gets a negative score. The greatest average sentiment is for Utilities at 0.176. Among the sectors, the lowest is Materials at 0.091 which is just over half of Utilities level. The ranking from highest to lowest is Utilities (0.176), Real estate (0.125), Tech (0.119), Energy (0.111), Healthcare (0.109), Industrials (0.107), Consumer (0.101), Communications (0.100), Financials (0.097), then Materials (0.091)

This gap can be explained by looking at the headlines for each industry. Both Utilities and Materials have really bad news stories, with earnings disappointments and regulatory stress. So it is not that one sector is getting a 'calm' and the other is getting a 'dramatic' coverage. However the difference is in the depth of the bad news. Materials worst headline ratings were -0.80, and the worst Utilities got was -0.62. To conclude, Materials still carries a heavier negative tail overall, its worst headlines are consistently more severe than Utilities.

Limitations. Since the data is headlines, rather than the full article, the sentiment index is essentially a noisy proxy for actual market sentiment. A single short headline often lacks the proper context needed to judge tone accurately, and the high neutral rate observed here is a partly reflection of that shortness as well as of VADER's vocabulary. Therefore a sentiment score of zero should not be read as 'no information' because it could simply be a neutral headline or one that the financial sentiment VADER could not detect.

4. Extensions & Innovations

Fund breadth. The minimum requirement for the brief is to create a single combined fund using two optimisation methods. Effortless Invest offers nine funds: three optimization methods (min variance, max sharpe, and risk parity), each is then applied across three asset universes (combined, equity only, and crypto only). This directly addresses the innovation part of the brief where stated: 'a wider or newer set of funds or optimisation methods than the required minimum'. This is the reason why the diversification benefit finding in section 2 becomes obvious, where nine funds exist in which to compare it, not one.

Sentiment Fusion. The main extension is a sentiment tilt on equity_min_variance. The optimisation step itself remains the same as the baseline fund, as these are just standard minimum variance weights computed without taking expected returns into account. The tilt only happens afterwards, and the weights are then re-scaled for each stock by a factor of 1 plus the strength times the sentiment score of that stock, with the sentiment score for each stock being the same 1 day lagged ticker sentiment used in Section 3. Not to mention, a stock with a positive sentiment will scale up and with negative sentiment will scale down. The scaling factor is then floored at 0.1, which means even if the sentiment readings is negative, the multiplier cannot go below 0.1, and the weights are normalised to add up to 100%. The strength parameter determines how important a sentiment is, where a strength value of 0.5 and a stock with a sentiment score of 0.2 will be assigned an additional 10% weight. The same stock at strength 2.0 will get a 40% additional weight boost. The tilt is applied after the optimisation process, so it does not risk any look ahead bias, as the sentiment signal was already lagged safely in Section 3.

Results. Table 2 and Figure 6 compare the base equity_min_variance fund against two tilt strengths. The effect is small, but it moves the same way across both metrics that matter. Return ticks up slightly, from 10.81% to 10.90% and volatility ticks down slowly from 15.98% to 15.89%. As for sharpe, it improves step by step as strength increases (0.676 to 0.678 to 0.686). Max drawdown gets worse at high strength, from -20.32% to -20.46%, which is a tiny trade off against the sharpe gain. This isn't a significant effect, but a real and consistent one. Tilting towards the stock with positive sentiment and away from stocks with negative sentiment pushes the fund towards a slightly better risk adjusted returns, without changing its overall risk.

Why is the effect small? This modest improvement isn't surprising given what Section 3 found. 48.8% of headlines score exactly neutral under plain VADER, meaning nearly half the ticker days feeding into the tilt have no signal at all. A tilt built on a signal that is neutral almost half the time can only nudge the portfolio so far, regardless of how non neutral the other half is. Therefore this links back to the sentiment index's own limitation directly to the fusion result, where a noisier more neutral heavy signal produces a smaller and less noisy tilt effect.

What wasn't pursued? A finance specific lexicon extension to VADER, as it's one of the brief's suggested innovation paths. Although it was considered, it was not built. Extending VADER's vocabulary to properly score finance specific language would most likely reduce the 48.8% neutral rate found in Section 3. Yes, it would have strengthened the fusion tilt's effect, as the tilt is directly restricted by how often the underlying signal carries real information. Given the scope of this project, this is left as a natural next step rather than a completed extension, an idea discussed further in Section 6.

5. The App & Investor Journey

Effortless Invest is designed for investors who want to diversify their investment into both equity and cryptocurrency markets. It's a tool that lets investors manage their investment systematically, rather than having to watch individual tickers or read financial news updates on its own. The app provides this via a streamlit dashboard organised around three tabs, Funds, Sentiment, and Data, with the Funds tab open by default

Landing and Fund comparison. Upon opening the app, the user will first see the product's title "Effortless Invest – Systematic Multi-Asset Funds" with a one line caption describing the product, and a fund comparison table. There is no welcome screen in between. All nine funds are presented at once on the table, this makes it easy for the user to compare each fund's annualised return, annualised volatility, Sharpe ratio, and maximum drawdown.

Fund Fact Sheet. Upon clicking on any specific fund, its fact sheet opens. This includes a growth of \$1 line chart, four metric cards with the fund's headline metrics, and a current holdings table showing the top ten tickers by weight at recent rebalance. This is similar to the structure in the exhibits of Section 2, but serves a different purpose. Section 2 supports detailed statistical analysis for a reader evaluating the backtest, while the app's fact sheet is designed for a quick comparison for investors to choose between funds.

Allocation. A user can adjust their hypothetical allocation by using the slider feature (0-100%) for each fund. The app will then show a growth of \$1 chart that represents that allocation. This is only a representation, not an actual investment decision. If the user were to choose a combination of a defensive equity and a higher returning crypto fund, it would show how it has performed compared to the other options, but not make any actual investments.

Sentiment. The sentiment tab is designed to show the sector sentiment index that was created in Section 3. This can be displayed by a multiselect filter to choose any of the 10 sectors. This provides the investor a market mood view in addition to the fund data without having to read the headlines themselves.

Deployment. The app is deployed on Streamlit Community Cloud from a public GitHub repository (private during development, public during submission), live at effortlessinvest-z5628210.streamlit.app. Following the brief's requirement to keep the app light, it only shows previously generated results and does not recompute anything live. Also, it does not run any backtest or sentiment scoring calculator itself. This is to ensure that the app is fast and reliable on Streamlit's free tier, which is unable to run the full backtest or VADER scoring pipeline on demand.

Design. The app uses a custom navy and white colour theme set in `.streamlit/config.toml`. This changed the default red accents in streamlit to a brighter blue which is the colour of finance. This is an extension of the design system introduced in Part A, where every report figure was designed with a consistent title format and borderless style. The app's theme extends that same design purpose to the product's front end, this shows that visual design

was treated at a deliberate and consistent choice across both parts of the project rather than an afterthought for just the app.

6. Critical Reflection & Recommendations

What worked well. The project delivers well via the brief's requirement of one combined fund with two methods. Nine funds in three asset universes, all optimised accurately, and no look ahead bias with the appropriate 252/365 day annualisation applied based on the calendar of the underlying asset. This breadth was the reason why the findings in Section 2 was possible, without the equity only and crypto only funds to compare, the diversification benefit would not have been seen at all. The sentiment index built on plain VADER itself confirmed the brief's own stated expectation where roughly half of finance headlines score neutral under a general purpose sentiment tool. This is an actual piece of evidence, not just an assumption. The app itself is also fully developed and functional, reading only precomputed results, not live computation. Lastly, it also extends the custom design system introduced in Part A into the product's front end.

What didn't work well. The sentiment model's limitation is also the one I have the clearest number for: 48.8% of headlines carry no measurable sentiment under plain VADER. This shows how limited the fusion tilt could shift portfolio weights. Also, no transaction costs were modelled anywhere in the backtest. This overstates every fund's real world return, and it does not overstate them evenly. Max sharpe and risk parity produces less stable weights between rebalances than min variance. Therefore they face higher real trading costs than the frictionless backtest shows. Not to mention, the 252 day window and monthly rebalancing were a reasonable and standard choice but they were never tested against alternatives. Hence, their impact on the result is not validated.

Recommendations.

- 1. Build a finance specific sentiment lexicon.** Plain VADER scores 48.8% of headlines as neutral, which capped the fusion tilt effect in Section 4. This is nearly half the ticker days feeding it that carries no usable signal. Therefore extending VADER's word list would cut this false neutral rate; this would give the fund a stronger signal to actually respond to sentiment on, instead of just leaving weights unchanged on nearly half of all trading days.
- 2. Add a transaction cost model.** All nine funds are backtested without transaction fees, this inflates actual returns. This effect isn't the same size for every method, max sharpe and risk parity shift their weight more between rebalances than min variance, hence they would face higher real trading cost than the backtest shows. Therefore, a simple cost per trade would make the fund ranking reflect what an investor would actually expect to earn.
- 3. Add a maximum single asset weight constraint for crypto funds.** Crypto_min_variance puts 48.5% in BTC and 24% in TRX, that's 72.5% of the fund in just two of the 10 available coins. The optimiser is technically doing its job by finding the lowest risk combination. In the real world, an investor or fund manager would still see this level of concentration as risky, no matter how calm the two coins have

looked historically. Therefore, a maximum weight cap per asset like no coin above 25-30% would force real diversification across the crypto universe. That trades a small amount of theoretical efficiency for much lower concentration risk .

Reference List

FINS3645 Course Team 2026, *FinTech Project 2026 dataset* [equity prices, crypto prices, news headlines], data set, UNSW Sydney, School of Banking and Finance.

Hutto, CJ & Gilbert, E 2015, (PDF) vader: A parsimonious rule-based model for sentiment analysis of Social media text, ResearchGate, January, viewed 24 July 2026, <https://www.researchgate.net/publication/275828927_VADER_A_Parsimonious_Rule-based_Model_for_Sentiment_Analysis_of_Social_Media_Text>.

Appendices

Figure 1

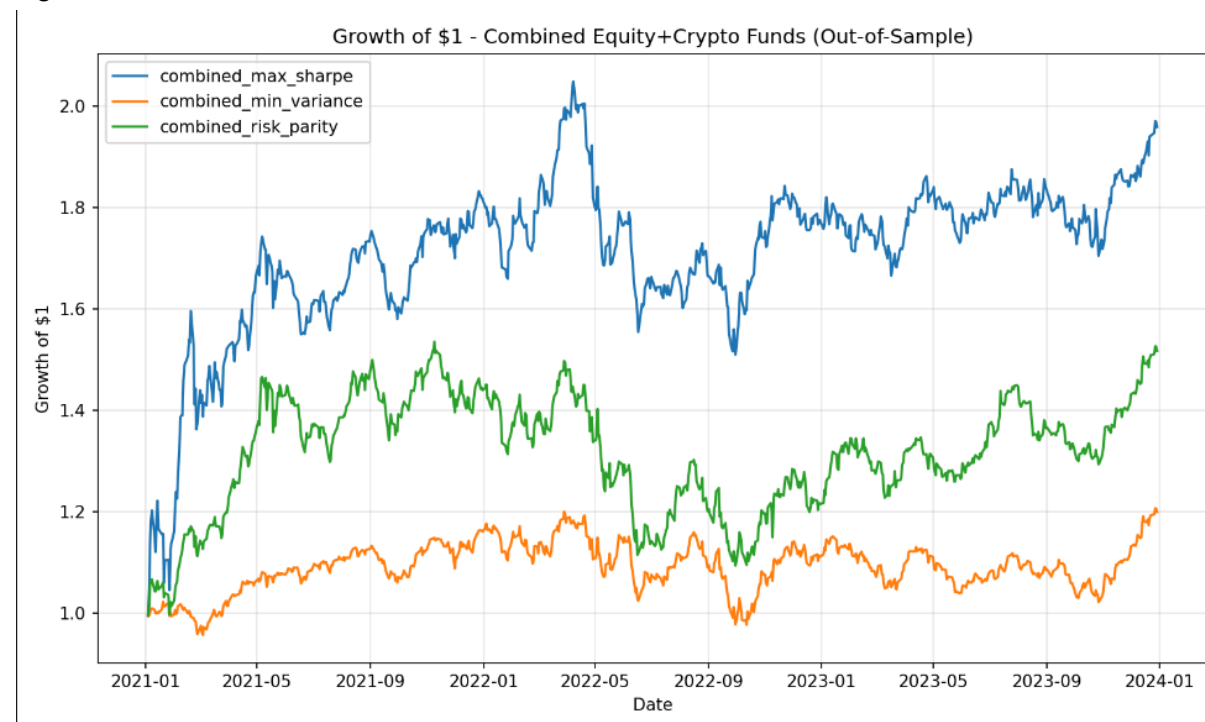


Figure 2

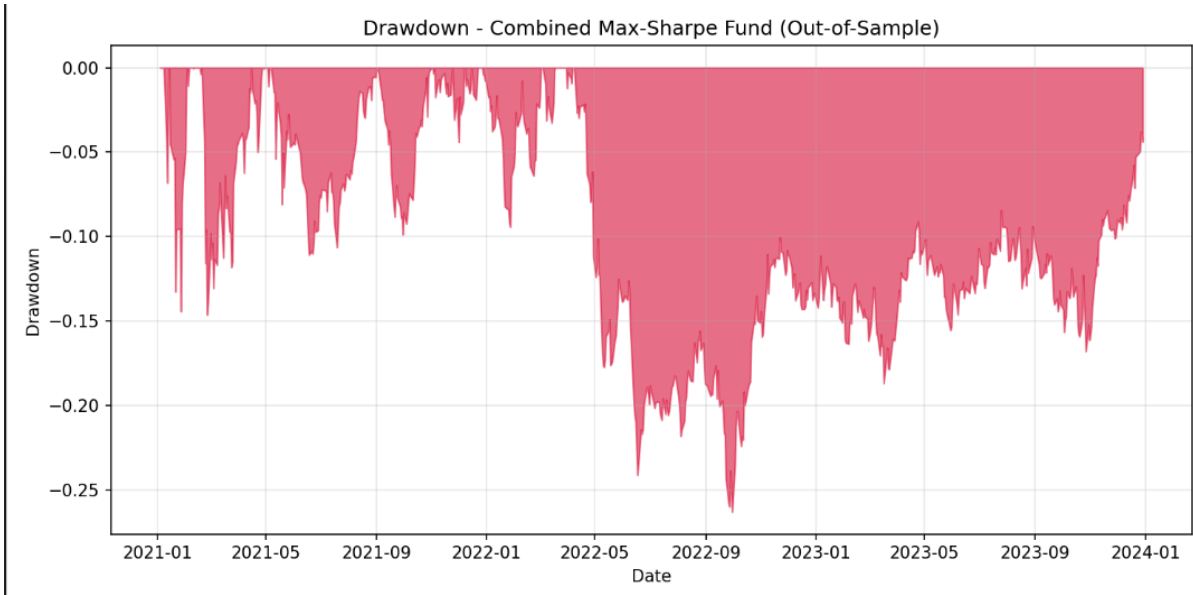


Figure 3

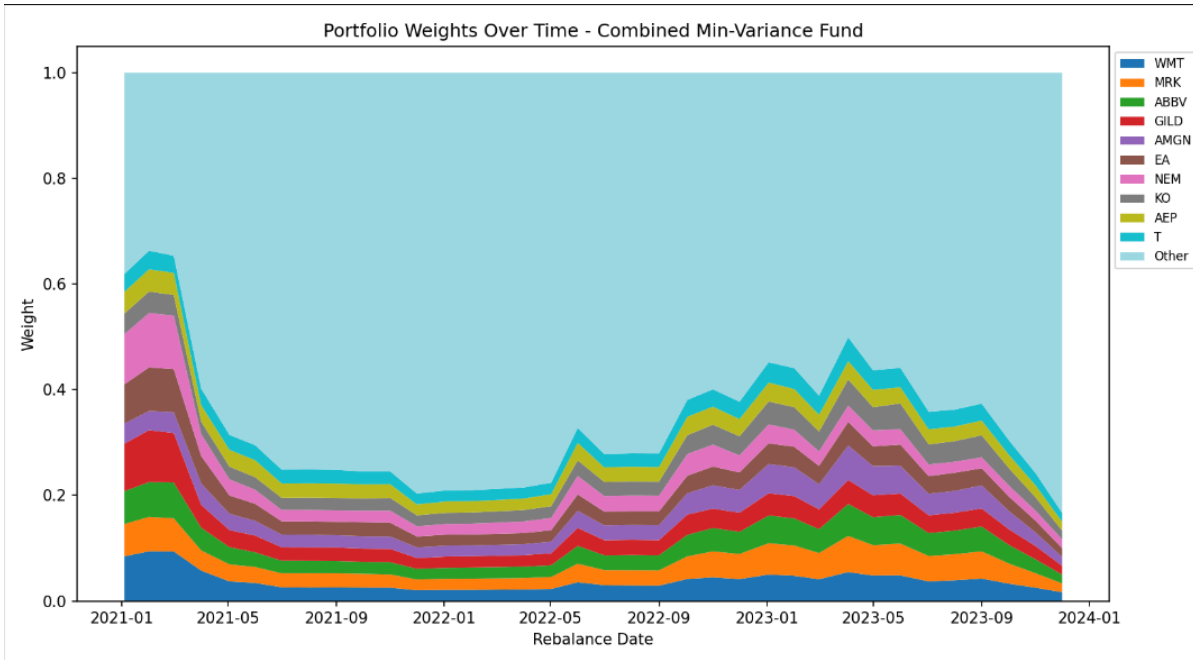


Figure 4

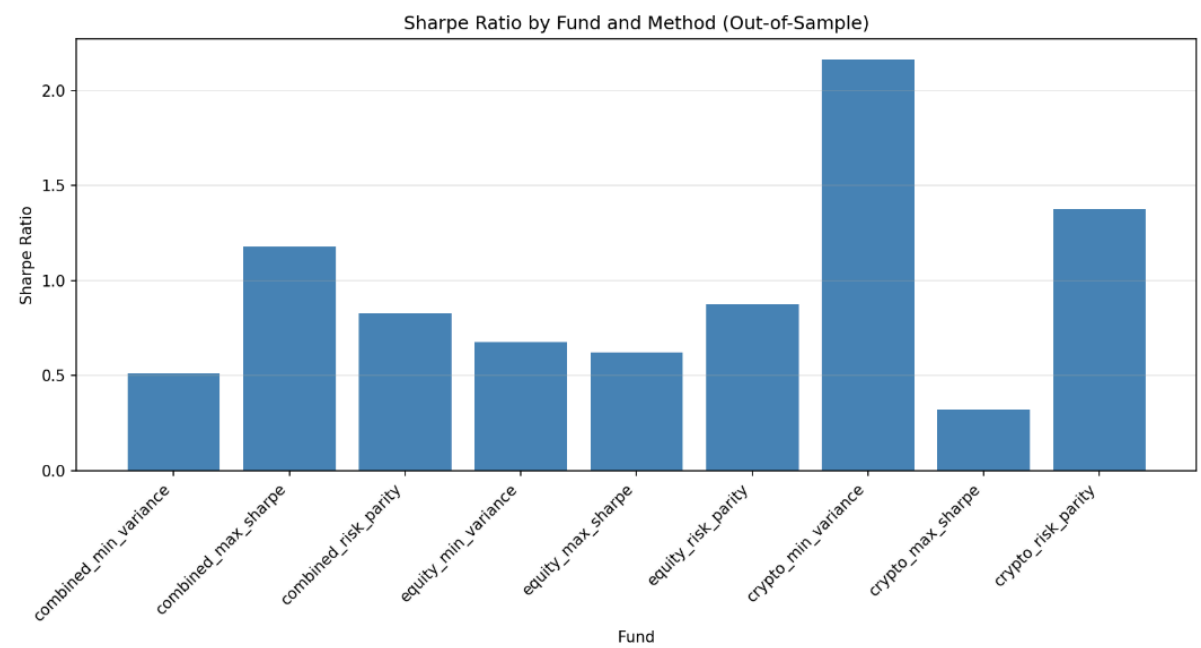


Figure 5

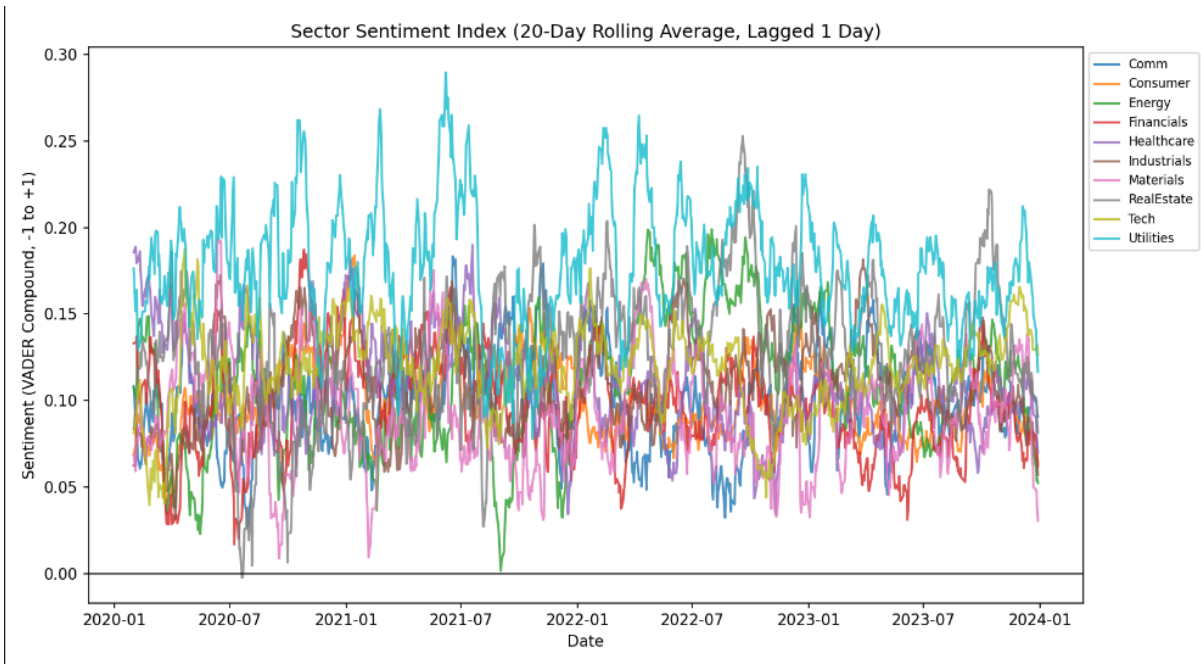


Table 1 (Performance Summary)

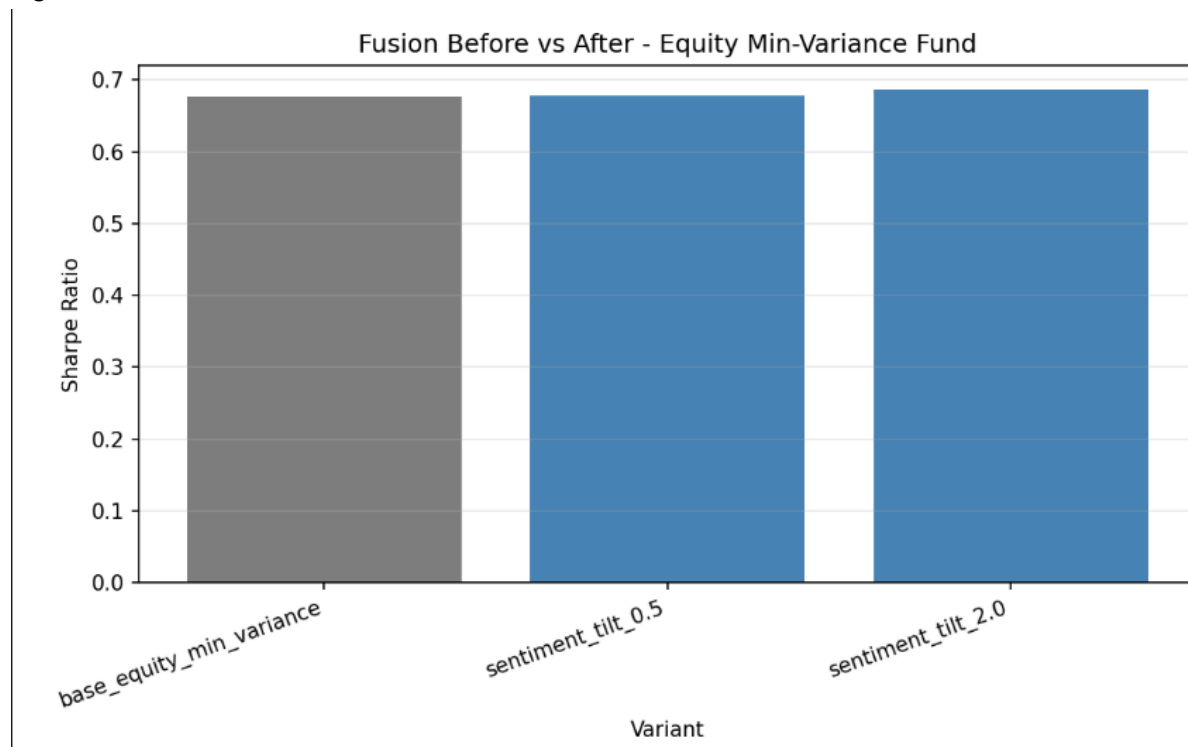
Fund	Ann Return	Ann Vol	Sharpe	Max DD
combined_min_variance	7.4%	14.5%	0.512	-18.6%
combined_max_sharpe	29.1%	24.7%	1.179	-26.3%
combined_risk_parity	17.6%	21.3%	0.828	-28.7%

equity_min_variance	10.8%	16.0%	0.676	-20.3%
equity_max_sharpe	11.3%	18.2%	0.623	-26.1%
equity_risk_parity	14.1%	16.2%	0.874	-20.3%
crypto_min_variance	146.7%	67.8%	2.164	-67.5%
crypto_max_sharpe	25.0%	77.4%	0.323	-89.3%
crypto_risk_parity	111.4%	81.0%	1.376	-81.6%

Table 2 (fusion_comparison)

Variant	Ann Return	Ann Vol	Sharpe	Max DD
base_equity_min_variance	10.81%	15.98%	0.676	-20.32%
sentiment_tilt_0.5	10.81%	15.95%	0.678	-20.35%
sentiment_tilt_2.0	10.90%	15.89%	0.686	-20.46%

Figure 6



Deployment Links

Live app: <https://effortlessinvest-z5628210.streamlit.app>

Public repository: https://github.com/shienzern/z5628210_projectB